



CS683: Advanced Computer Architecture-2024

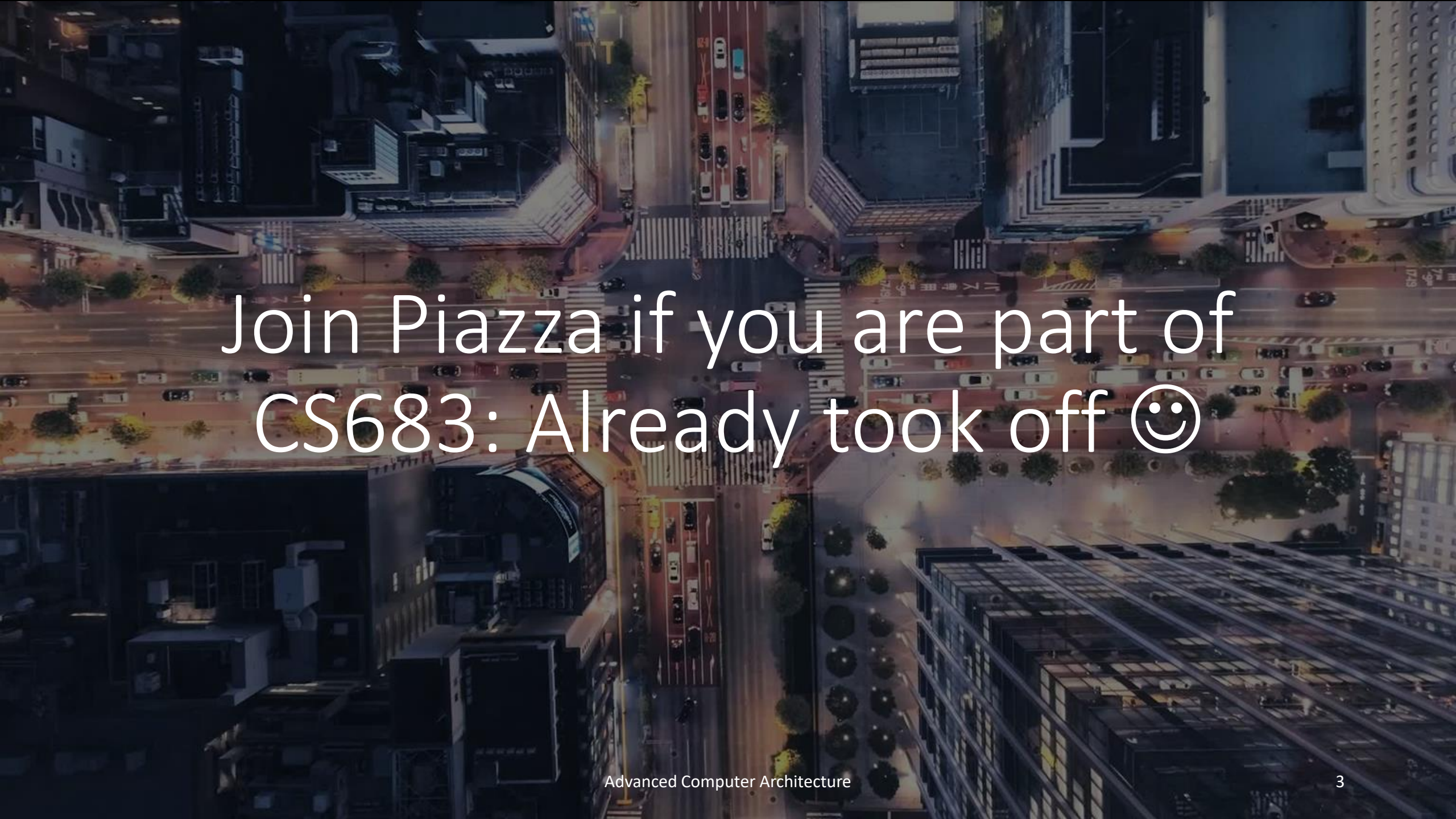
L4/L5-Welcome to Microarchitecture!!

<https://www.cse.iitb.ac.in/~biswa/courses.html>

<https://www.cse.iitb.ac.in/~biswa/>



Phones (smart/non-smart)
on silence plz, Thanks

An aerial night view of a city intersection. The scene is illuminated by streetlights, showing a grid of roads with cars and crosswalks. Tall buildings line the streets, and the overall atmosphere is urban and vibrant.

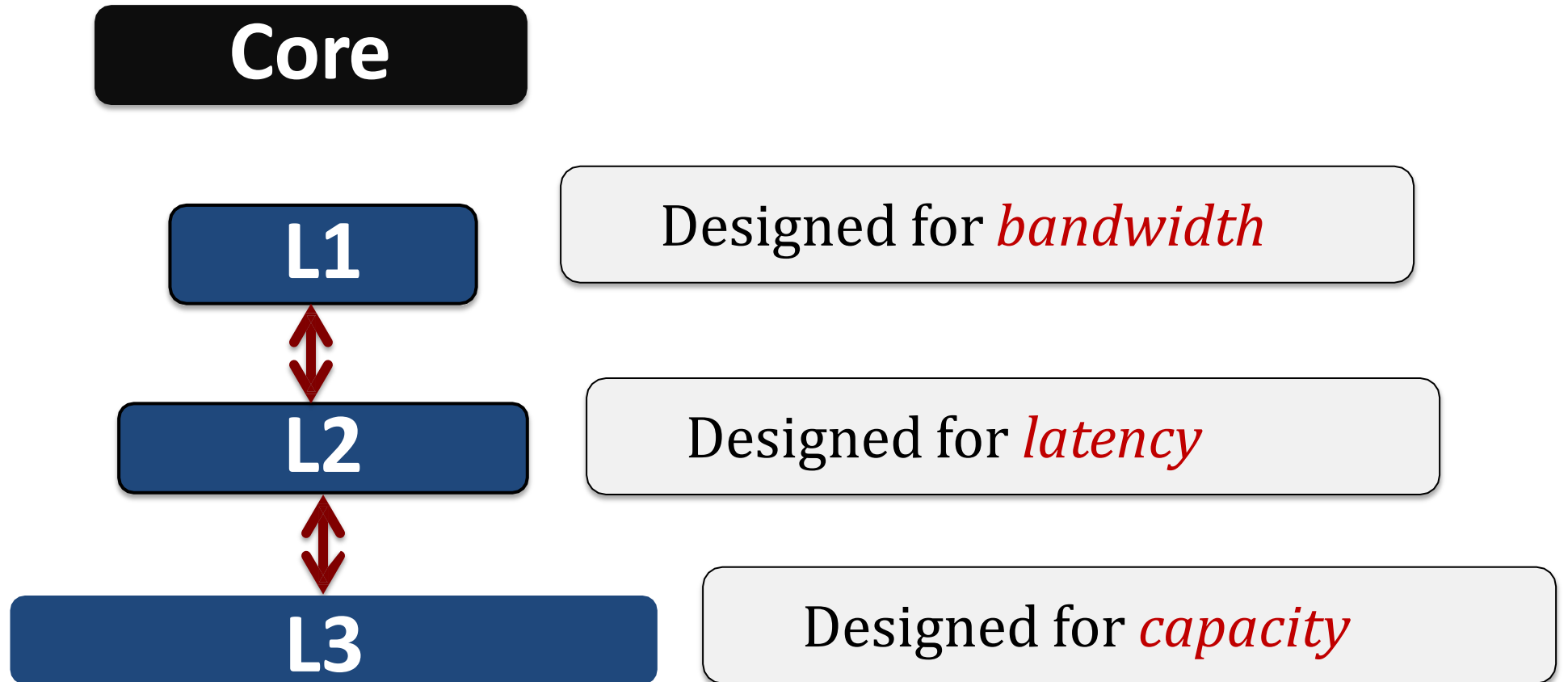
Join Piazza if you are part of
CS683: Already took off 😊

Let's understand some terms

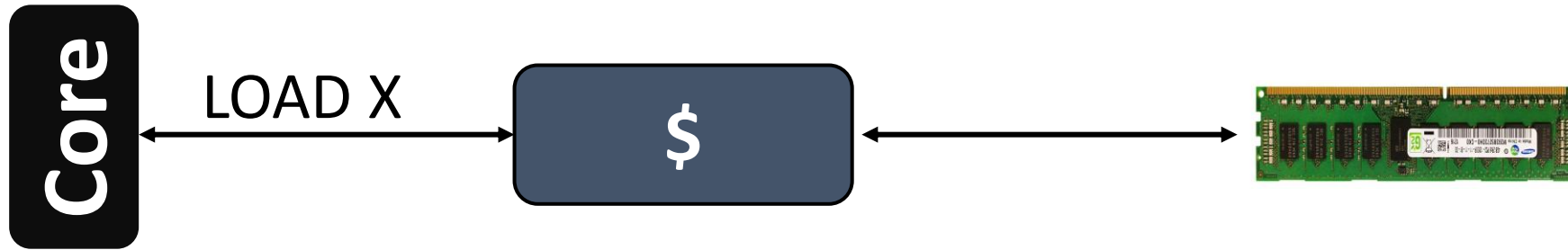
Hit, Miss, Miss penalty

Sequence of events in a typical cache hierarchy

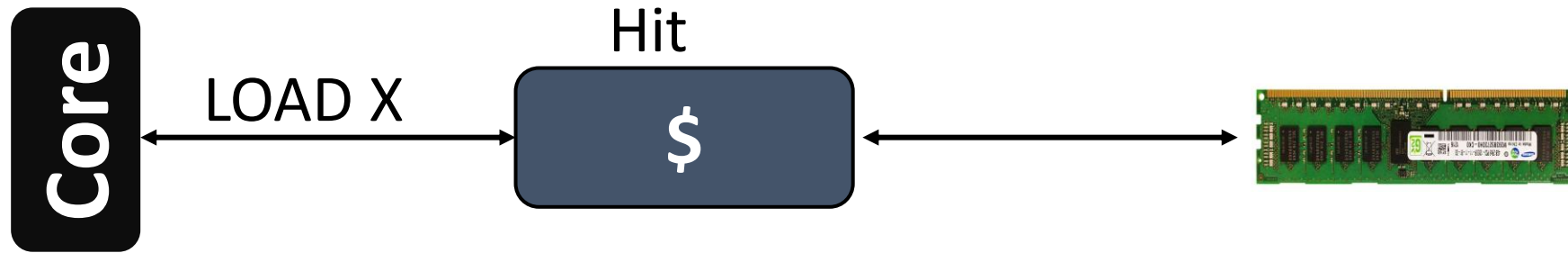
Cache Hierarchy



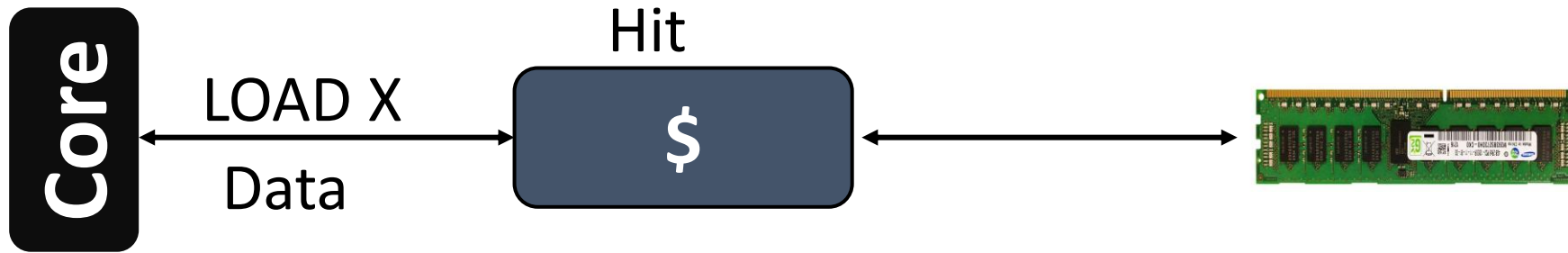
Core, Cache, And DRAM Interaction



Core, Cache, And DRAM Interaction

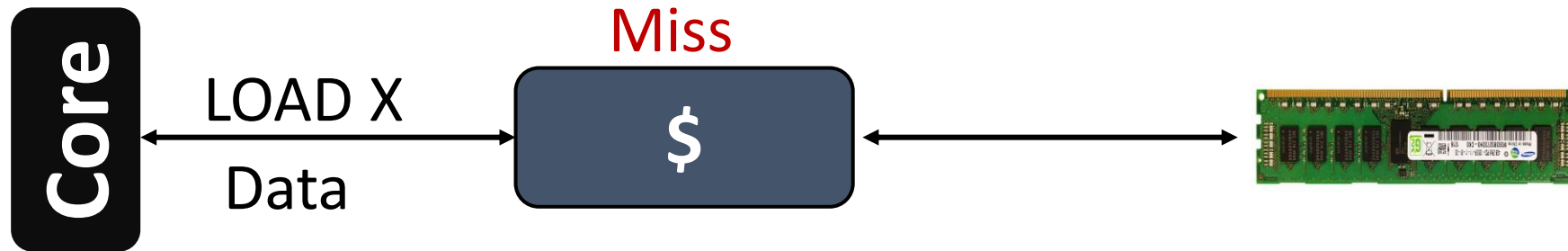


Core, Cache, And DRAM Interaction



Few cycles

Core, Cache, And DRAM Interaction



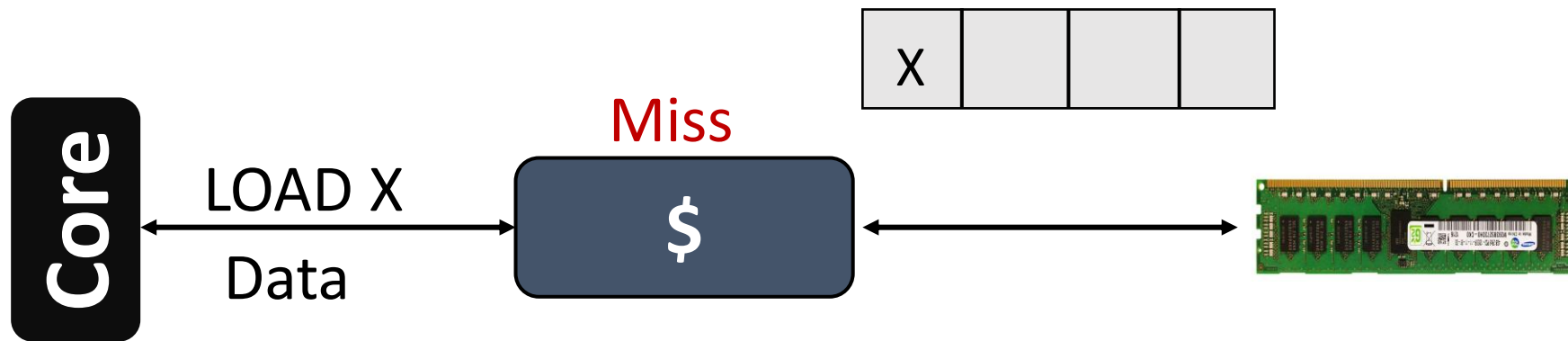
100s of cycles

I am an out-of-order core

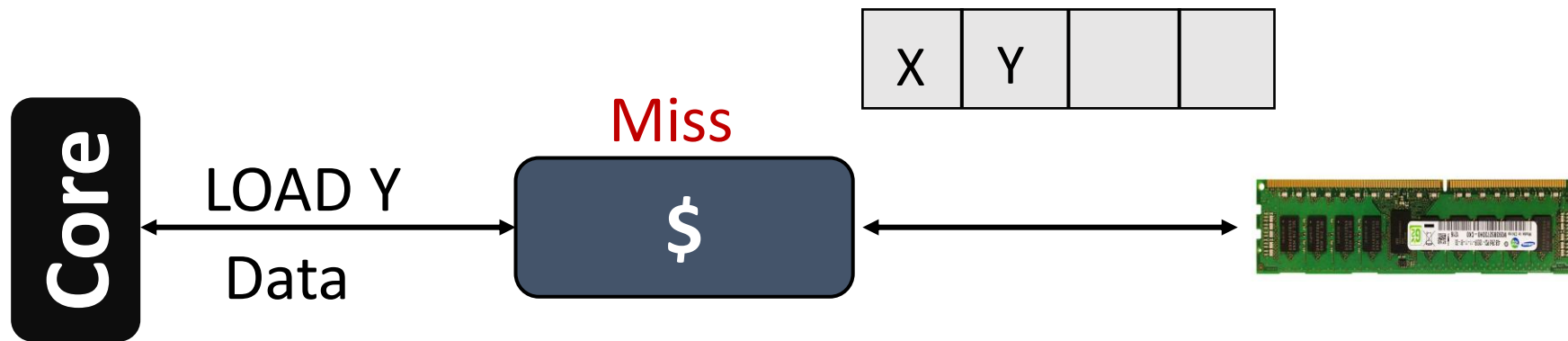


One cache miss and can't handle anymore misses

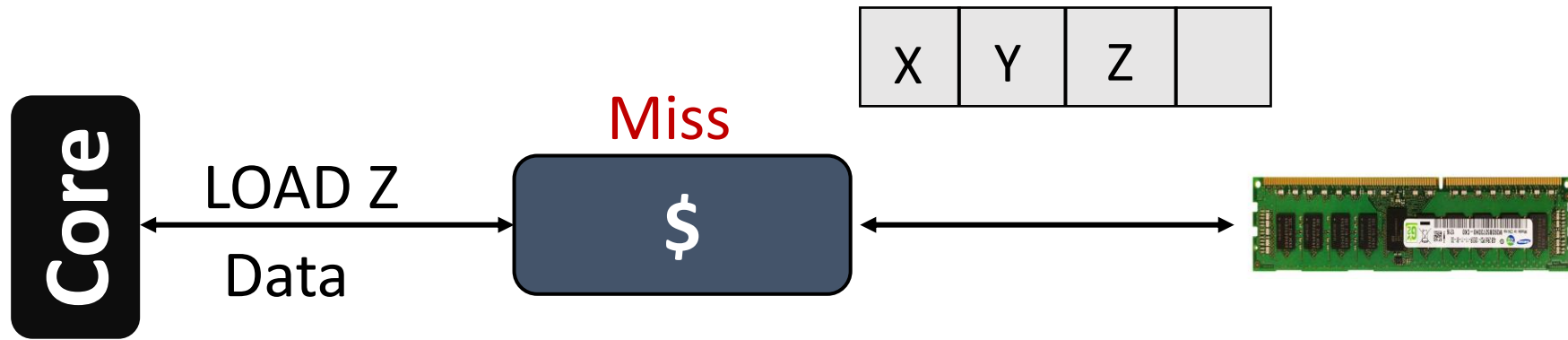
World of MSHRs (Miss Status Holding Registers)



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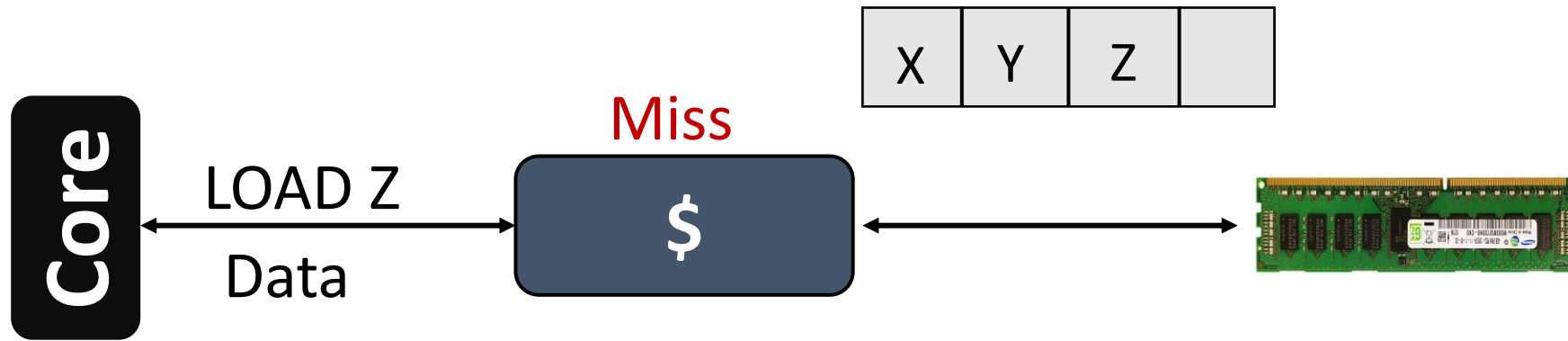


World of MSHRs (Miss Status Holding Registers)



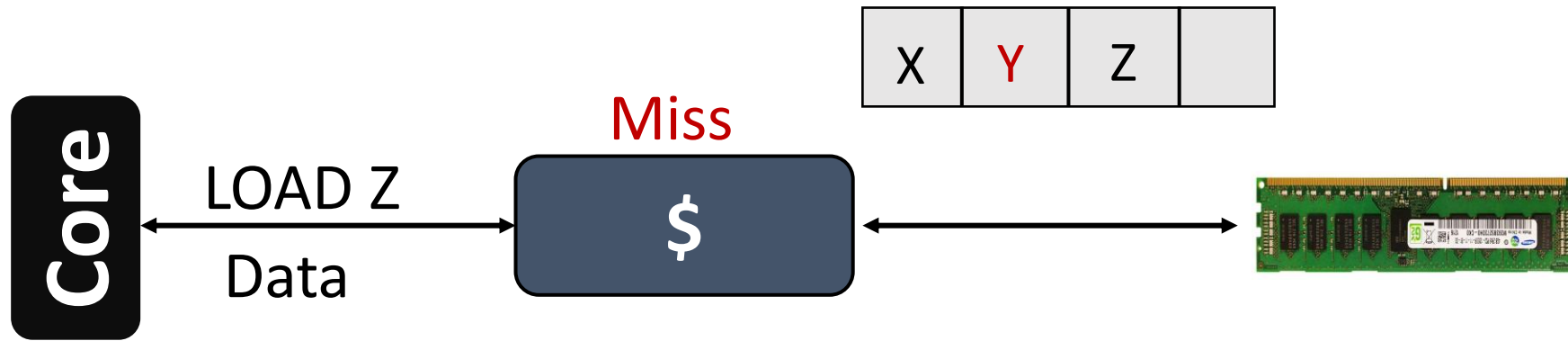
K-entry MSHR allows K outstanding misses from outer levels of memory hierarchy provides memory-level parallelism

World of MSHRs (Miss Status Holding Registers)



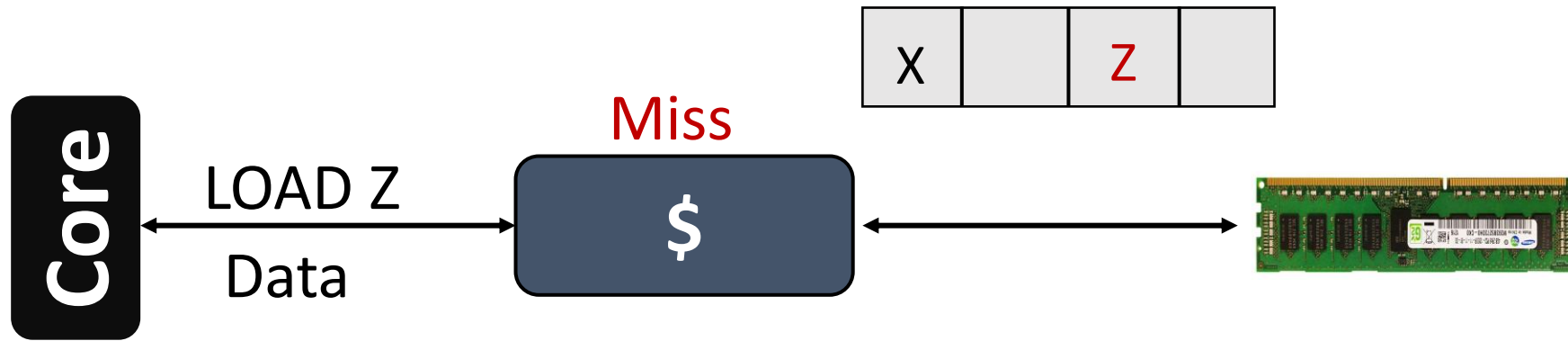
DRAM response time is not constant: can take from 60 cycles to 1000s of cycles (on a multi-core system).

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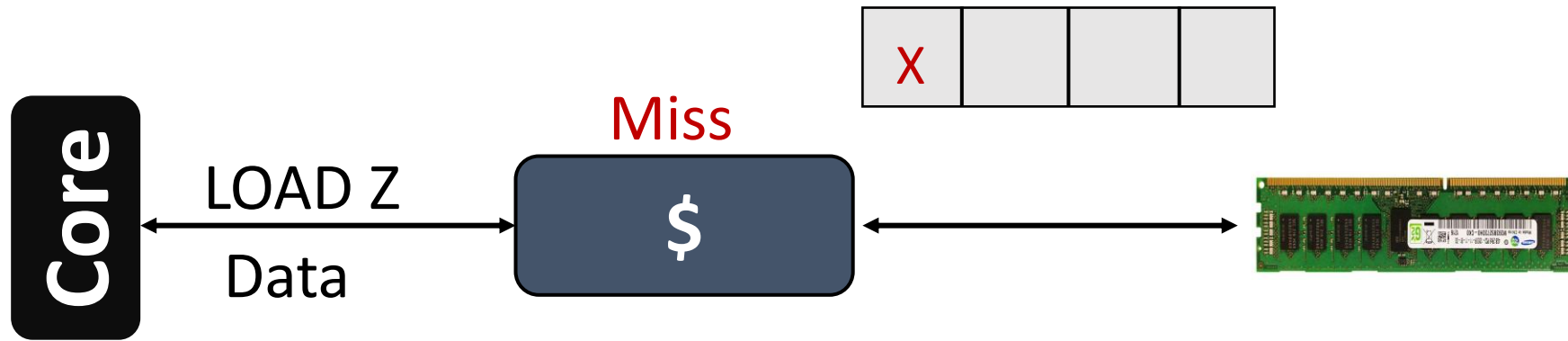
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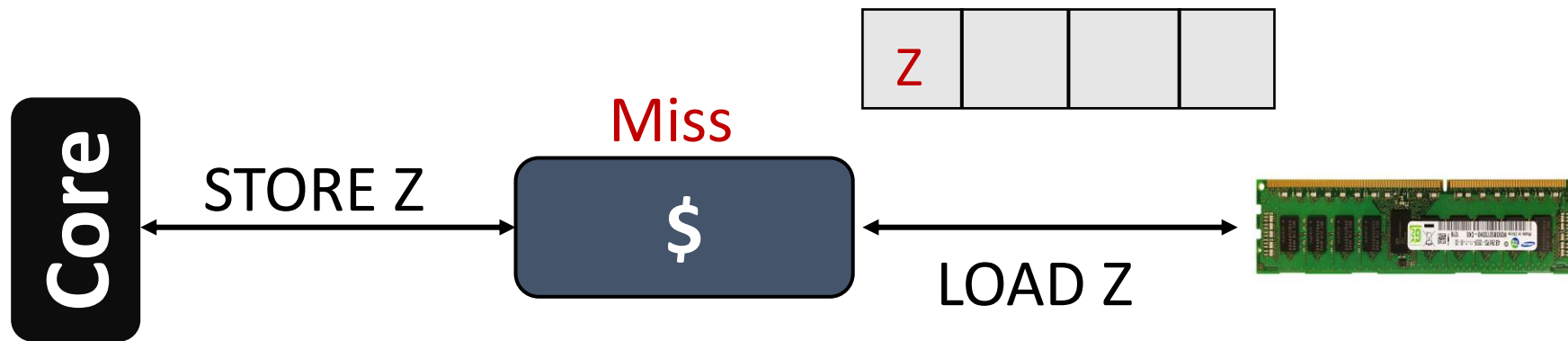
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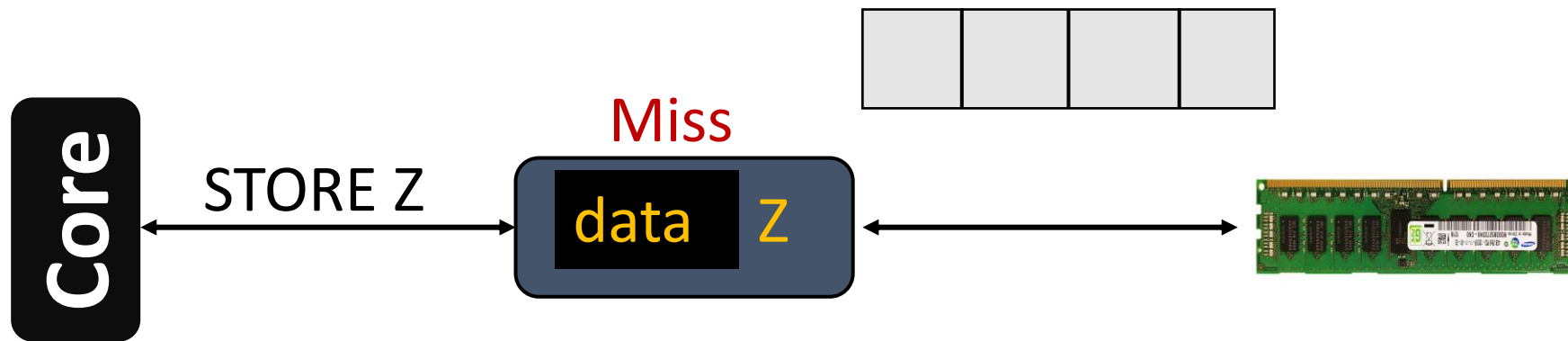


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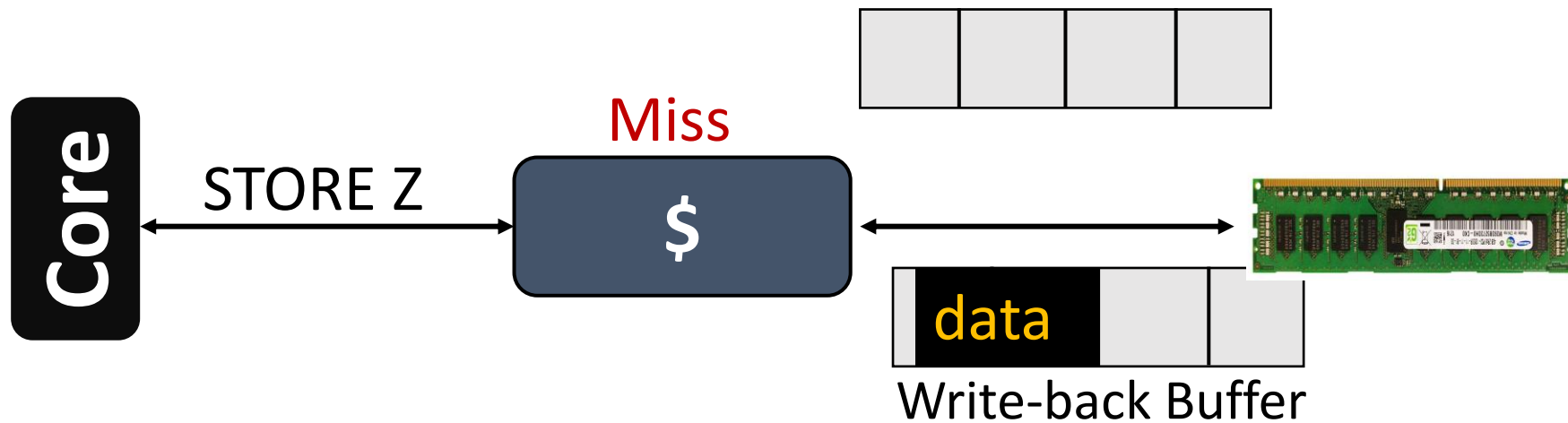
What About Writes (STOREs)?



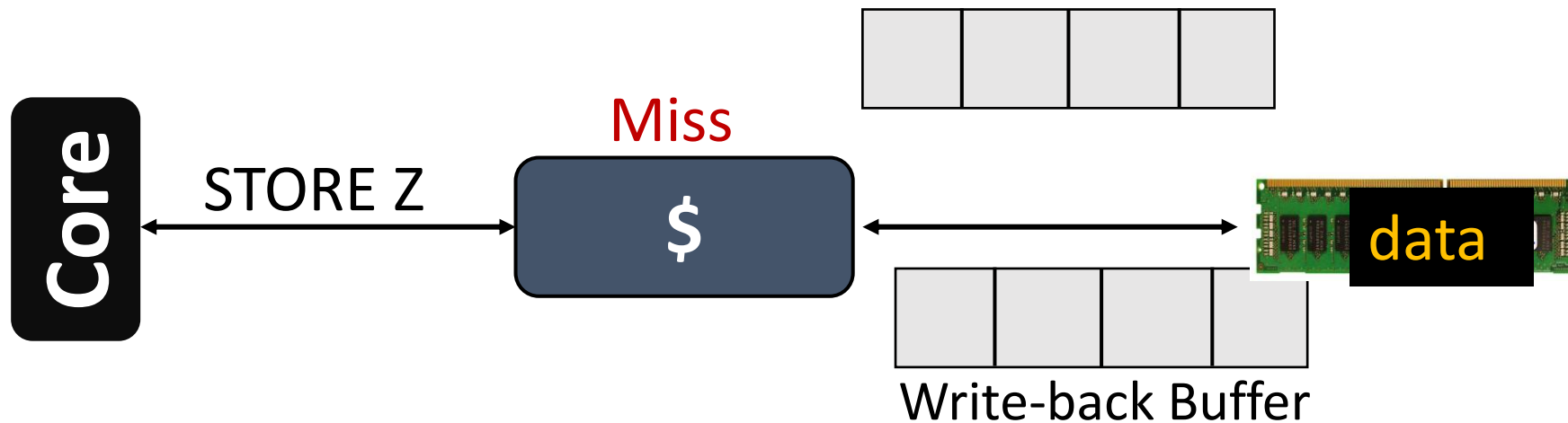
What About Writes (STOREs)?



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Write-back cache

In general, STOREs are not critical for performance. But why?

Blocking and non-blocking cache

- One miss can block the processor
- MSHRs in action (Cache becomes non-blocking)

Multiple misses that are outstanding at a given cache level.

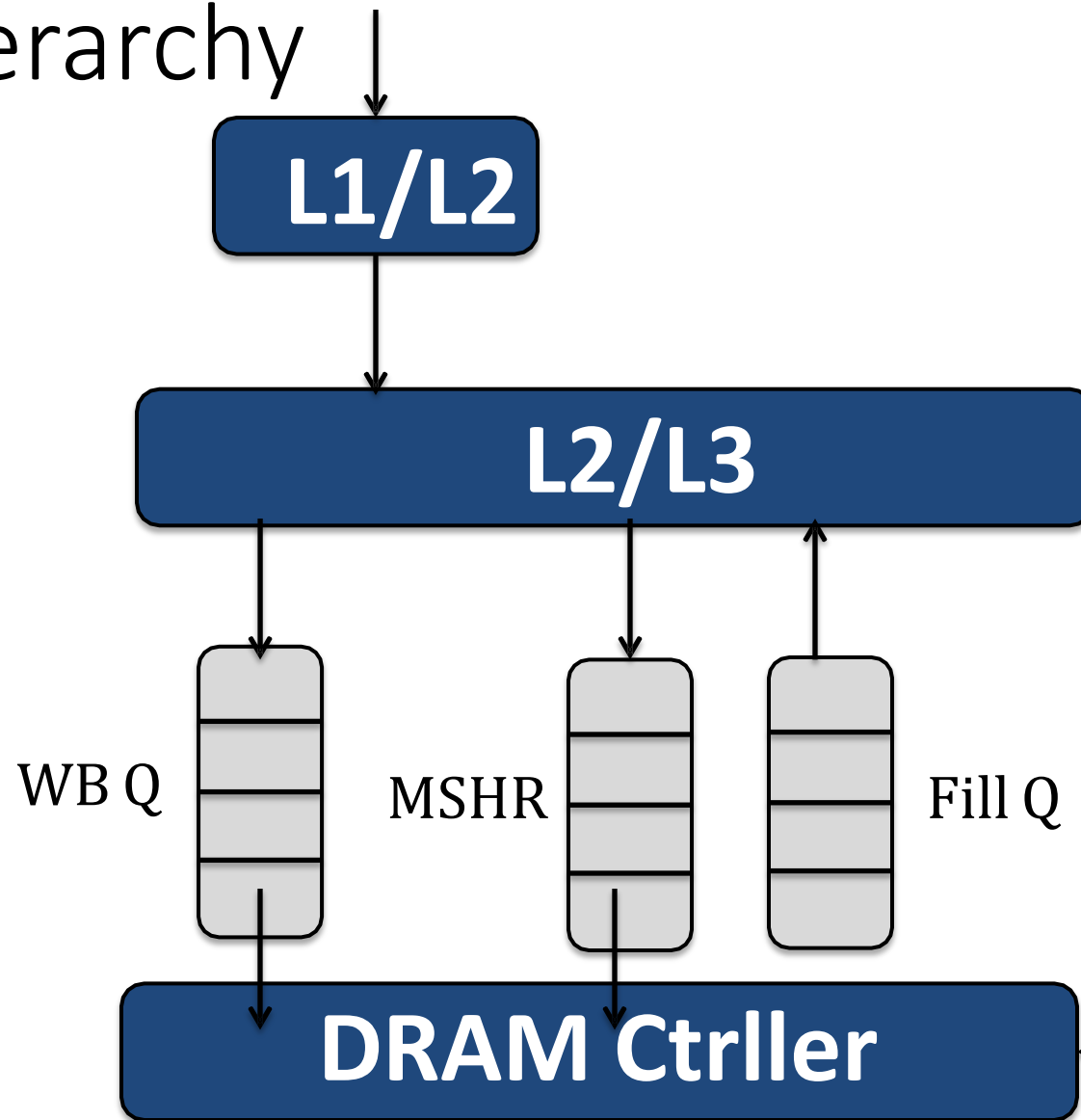
Helps in providing memory-level parallelism (MLP)

Helps in merging multiple requests to one cache line aligned address

MSHRs in the hierarchy

- If L1 has K entries then all the outer levels should have at least K entries to make it complete non-blocking cache hierarchy.
- Note that the responses coming from outer levels of cache and DRAM is out of order.

Cache Hierarchy



The complete picture

- Fill buffer: For storing the outstanding data coming from outer levels of caches.
- MSHRs: For storing outstanding misses
- Writeback q: For storing pending writeback requests to the next level.
- All these structures are used between two levels of the memory hierarchy

Cache hierarchy contd.

Inclusive: L1 cache blocks are present in LLC

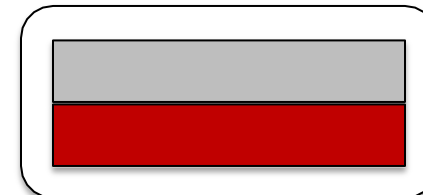
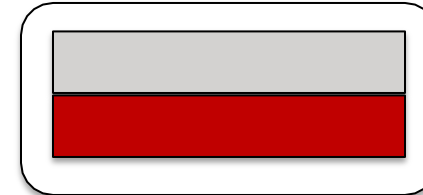
Exclusive: L1 cache blocks are absent in LLC

Non-inclusive: L1 cache blocks may/may not be in LLC

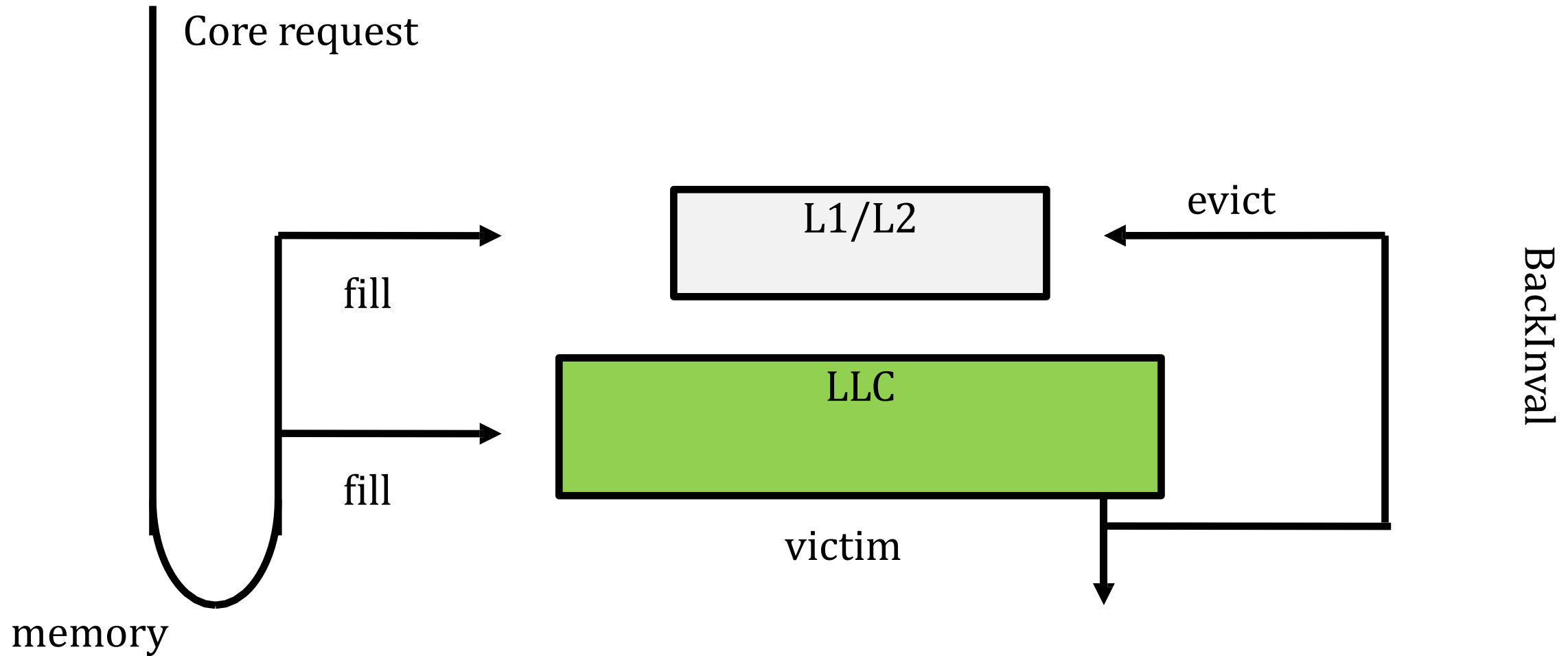
L1



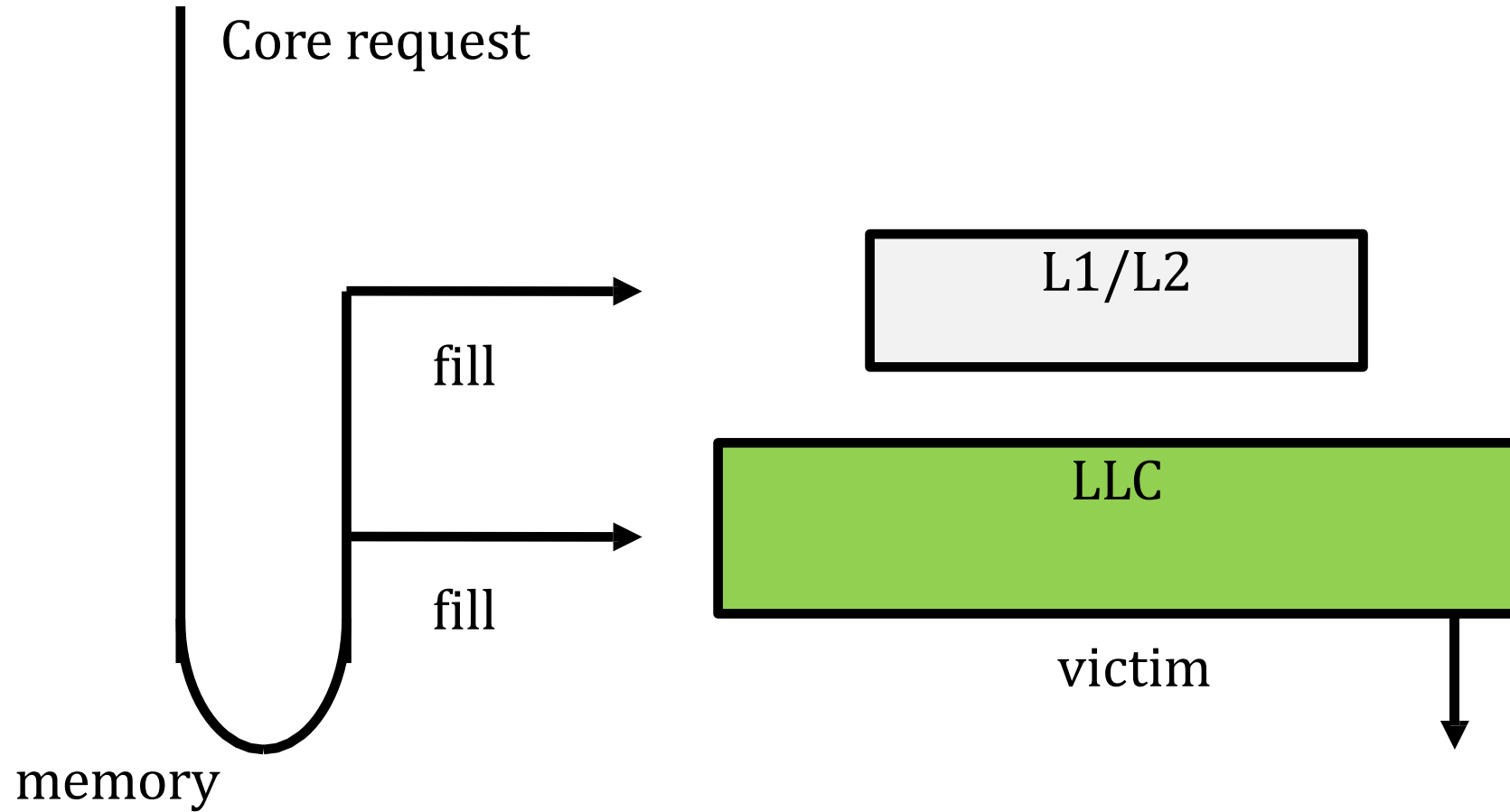
LLC



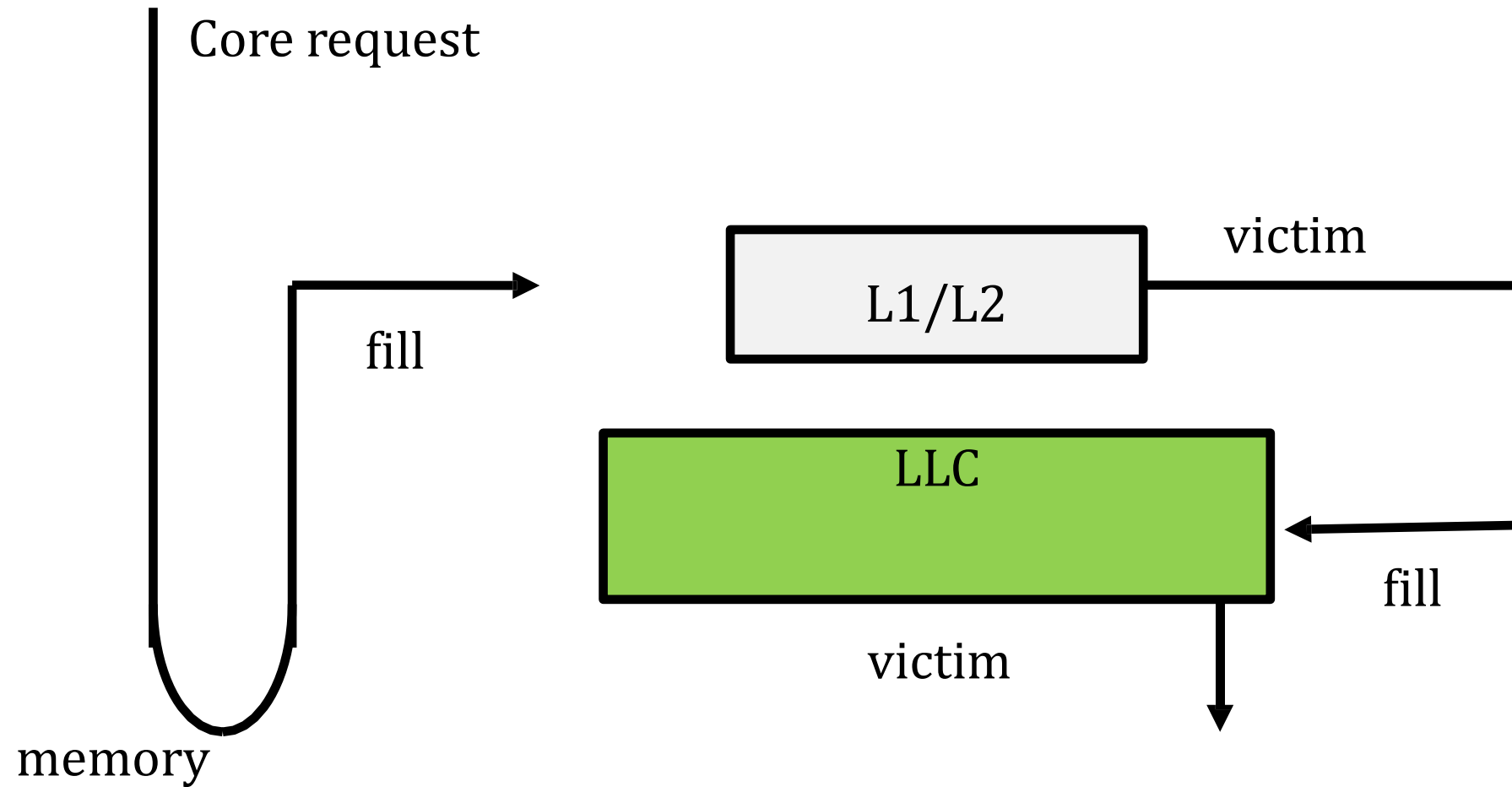
Inclusiveness



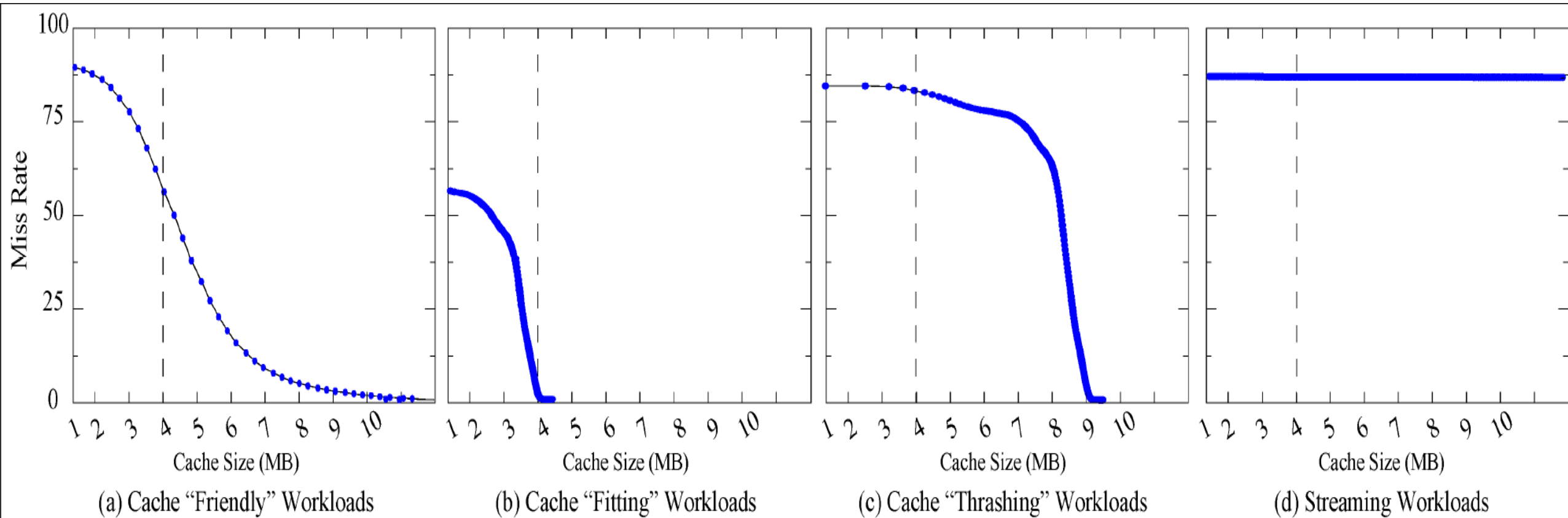
Non-inclusive (Commercial machines)



Exclusive



Types of applications



Logistics

- Mini quiz on August 23